

Suggested FYP topics for 2026-27

by

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Project code :	GLX1
Project title :	Elder Care Live Monitoring Dashboard
No. of students :	2
Description of project :	Context As the global population ages, the demand for elder care services is rising rapidly. Monitoring the well-being of elderly individuals, particularly those who live alone or in assisted living facilities, is crucial to ensure their safety and quality of life. Traditional monitoring systems often rely on manual checks or basic alert systems, which can be insufficient in detecting subtle changes in behavior that may indicate health issues or deteriorating conditions. Problem Statement The goal of this project is to develop a comprehensive live monitoring dashboard designed specifically for elder care. The system will utilize advanced ML algorithms and streaming to monitor and analyze the behavior and activities of elderly individuals in real-time while preserving their privacy. The project focuses on: 1. Privacy-preserving behavior monitoring using skeleton streaming and Human Activity Recognition (HAR). 2. Real-time analysis and alert generation for potential health or safety issues. Solution Approach and Tasks The primary challenge in elder care monitoring is ensuring accurate, real-time analysis while maintaining the privacy and dignity of the individuals being monitored. By leveraging recent advancements in skeleton-based activity recognition and avatar-based reconstructions, we can achieve this balance.

Key Project Steps:

- **Skeleton-based Behavior Monitoring:**

- Implement skeleton streaming technology to monitor elderly individuals' movements without capturing identifiable video footage, ensuring privacy is maintained.
- Develop and fine-tune Human Activity Recognition (HAR) algorithms to detect and classify daily activities, such as walking, sitting, or lying down, in real time.
- Integrate an alert system that triggers notifications for caregivers in case of unusual or potentially dangerous behavior patterns (e.g., falls, prolonged inactivity).

- **Avatar-based Reconstruction:**

- Utilize avatar-based reconstructions to visualize the monitored activities in a non-intrusive manner, allowing caregivers to assess the situation without exposing the elderly individual's identity.
- Explore the potential for integrating this technology with virtual reality (VR) environments for enhanced monitoring and caregiver interaction.

- **Data Integration and Analysis:**

- Combine data from various sources, such as motion sensors, smart cameras, and wearable devices, to provide a comprehensive view of the individual's well-being.
- Develop machine learning models to analyze this data and predict potential health issues, enabling preemptive intervention.

- **Dashboard Development:**

- Design a user-friendly dashboard that displays real-time monitoring data, alerts, and historical activity logs tailored for use by caregivers and healthcare providers.
- Ensure that the dashboard is accessible on various devices, including smartphones, tablets, and desktop computers, to facilitate ease of use in different settings.

Team Structure and Responsibilities [if it's for a team]

	The project is designed to be completed by a team of 1-2 students. The roles and responsibilities are as follows: • System Architect: ○ Lead the development of the overall system architecture, focusing on the integration of various technologies and ensuring system scalability. • Algorithm Developer (AI Specialist): ○ Focus on developing and refining the Human Activity Recognition (HAR) algorithms and ensuring their accuracy and reliability. ○ Work on the integration of avatar-based reconstruction technologies. • Program Manager: ○ Oversee the project timeline, coordinate between team members, and ensure that milestones are met on time. ○ Manage communication with external stakeholders, such as elder care facilities and healthcare providers. References • Harmony: Heterogeneous multi-modal federated learning through disentangled model training
Industrial collaboration :	No.

Project code :	GLX2
Project title :	Efficient 3D Perception Fusion in Infrastructure-Assisted Autonomous Driving Systems
No. of students :	2
Description of project :	Context As autonomous driving becomes more popular, its safety issues are gaining more and more attention. Traditional single-vehicle autonomous driving relies on onboard sensors and has a limited perception range. In complex traffic, such as city street occlusion or multi-vehicle scenarios, blind spots will appear, bringing risks. By combining roadside infrastructure and vehicle data to assist autonomous driving, its perception range can be expanded. However, there are problems with current lidar-based point cloud

fusion. Large-capacity point cloud data requires high-bandwidth communication and high computational cost and relies on precise positioning, which is difficult to maintain under realistic conditions. In vision-based roadside camera 3D object detection, depth prediction methods face instability due to camera parameter changes. A new framework for predicting pixel-to-ground height shows better roadside adaptability.

Problem Statement

This project targets the real-time perception fusion problem in infrastructure-assisted driving. Existing point cloud fusion has data, cost, and positioning defects, while roadside depth-based detection is ineffective. The goal is to integrate a new height-based 3D object detection framework into the existing real-time perception system to replace the roadside detection solution. The integrated system should:

1. **Improve the accuracy and stability of roadside 3D detection and reduce sensitivity to extrinsic.**
2. **Optimize the fusion process, accurately and efficiently combine the data of both parties, and obtain more accurate results**
3. **Reduce data transmission and cost and improve real-time performance.**

Solution Approach and Tasks

Key Project Steps:

The main task is to integrate the new height-based framework into the real-time perception system, take advantage of the framework's roadside detection, solve data, cost, positioning, and other issues, and achieve high-precision, real-time fusion. The integrated system must work stably under various traffic conditions.

- **Framework Integration:**
 - Learn the existing BEV-based detection framework, including its encoder, depth estimation module, projection module, and other components.
 - Design and implement the perception fusion mechanism and algorithm that integrates the

	roadside and vehicle side (from rule-based to learning-based schemes). • Data Transmission and Processing: ○ Design data transmission format according to the requirements of the new framework and adjust it for real-time performance and bandwidth budget. ○ Design dynamic load balancing, such as priority scheduling, early stopping, and batch alignment • System Robustness and Verification: ○ Simulation experiments to verify the correctness of the system in different scenarios. ○ Verify the impact of different interference factors on system accuracy and analyze the system's ability to suppress interference. Team Structure and Responsibilities (if it's for a team) The project is designed to be completed by a team of 1-3 students. The roles and responsibilities are as follows: • System Architect: ○ Design the integrated system's architecture, focusing on module integration and scalability. ○ Decide on technology and solve integration-related technical problems. • Algorithm Developer (AI Specialist): ○ Optimize the new framework's detection algorithm and develop fusion algorithms ○ Implemented and debugged algorithms, ensuring they fit the system's architecture. • Program Manager: ○ Manage the project schedule, set milestones, and ensure timely completion. ○ Communicate with external parties and adjust the project plan as needed. References • VIPS: Real-time perception fusion for infrastructure-assisted autonomous driving
Industrial collaboration :	No.